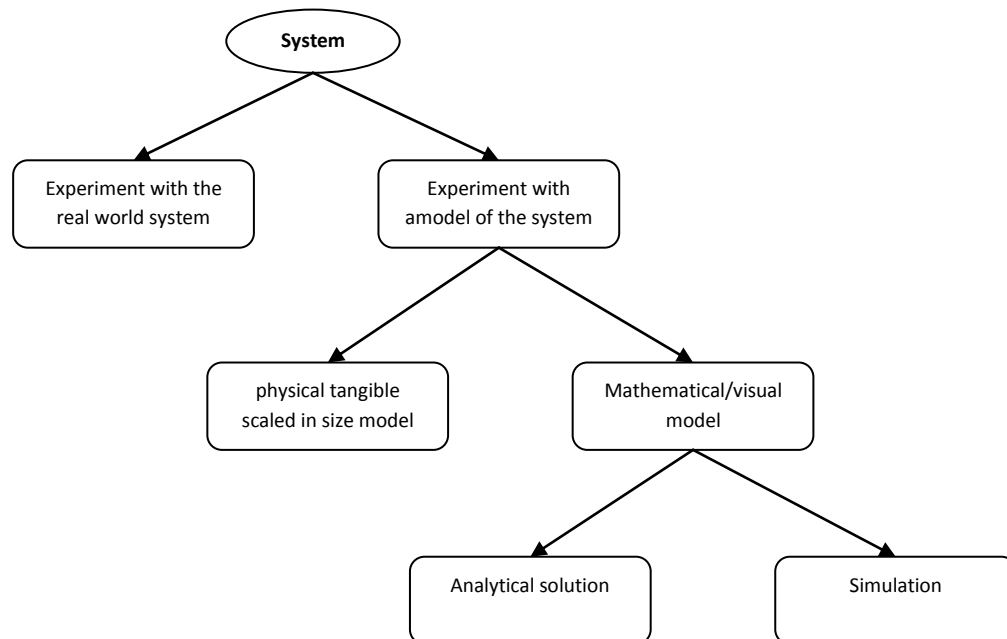


Reader 001

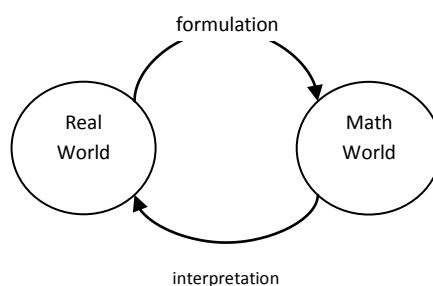


A cyclic process [**Mathematical Model**] is derived thus to address the primary premise;

Formulate: change problem into a form that allows use of some mathematical analysis

Solve: use mathematical rules to solve the problem that has been set up by transformation

Interpret: convert the mathematical solution back to the original form



Central to this process is the representation of real world problem by a mathematical structure such as a graph, an equation, or an inequality. Such representation is what is meant by a **Mathematical Model**.

FUNDAMENTALS

This section deals with some fundamental concepts that will be used during the discussion of this reader. A good comprehension would guaranty ease of learning in later part of the course.

System: A system is a group of objects that are joined together in some regular interaction or interdependence towards the accomplishment of some purpose.

System Boundary: System Boundary is a logical limbo line that defines the interaction point between the system internals and outside externals. Think of it as a demarcation of system scope or in a more physical sense system contours.

System Environment: A system is affected [observable change in state] by the changes occurring outside system. These changes are said to reside in system environment. Environment can be universal or existential depending on the constraints we imply on our model.

System Entity: An entity is an object of interest in the system.

Attribute: An attribute is a property of an entity. All entities have some attributes. These attributes have a symbolic representation which may reside in a defined set of established knowledge base.

State of Object: The accumulated value (continuous/discrete/qualitative) of all the attributes (variables) present in entities (objects of interest), represents state of that entity.

State of the System: The collection of variables (values of interest in entities) necessary to describe the system at a given time stamp, from a particular perspective (objective of investigation on system under study), is called state of the system at that time.

Activity: An activity is an action (exchange of data within system entities with a definite start and ending) that takes place over a period of specific and defined length of time and brings an observable change in the state of system.

Event: An event is an instantaneous occurrence (??) that might change the state of the system

Class Example Test:

System	Bank
Entities	Customers, Accounts
Attributes	Balance in an account, Interest Rate
Activity	Making a deposit
Event	Customer arriving at bank
Variables	Free tellers, customers in queue, arrival time of a new customer, time spend on serving one customer

Class Example Test:

System	Entities	Attributes	Activities	Events	Variables
Railway station					
Popcorn production unit					
Network communication					
Your dress wardrobe					
Our galaxy					